

Neural Networks - Practice Questions

Topic: Neural Networks

Q1. (Multiple Choice) [Easy]

What is a Neural Network inspired by?

- A. Computer processors
- B. The human brain and biological neurons
- C. Statistical equations
- D. Database systems

Q2. (Multiple Choice) [Easy]

Which of the following is NOT a layer type in a basic Neural Network?

- A. Input Layer
- B. Hidden Layer
- C. Output Layer
- D. Processing Layer

Q3. (Multiple Choice) [Easy]

What is the role of an Activation Function in a Neural Network?

- A. To initialize weights
- B. To introduce non-linearity so the network can learn complex patterns
- C. To calculate the cost function
- D. To split data into training and testing sets

Q4. (Multiple Choice) [Medium]

What is the ReLU activation function formula?

- A. $f(x) = 1/(1+e^{-x})$
- B. $f(x) = \max(0, x)$
- C. $f(x) = (e^x - e^{-x})/(e^x + e^{-x})$
- D. $f(x) = x^2$

Q5. (Multiple Choice) [Medium]

What is Backpropagation in Neural Networks?

- A. Forward pass of data through all layers
- B. Algorithm to update weights by propagating error backwards from output to input
- C. Process of initializing weights randomly
- D. Method to add more hidden layers

Q6. (Multiple Choice) [Medium]

What is a Convolutional Neural Network (CNN) primarily used for?

- A. Text generation
- B. Image and spatial data processing

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- C. Time series prediction only
- D. Clustering unlabelled data

Q7. (Multiple Choice) [Medium]

What problem does the Dropout regularization technique solve?

- A. Underfitting
- B. Overfitting
- C. Slow convergence
- D. Missing data

Q8. (Multiple Choice) [Hard]

What is the Vanishing Gradient problem?

- A. When gradients become too large during backpropagation
- B. When gradients become extremely small during backpropagation, making early layers learn very slowly
- C. When the model fails to converge at all
- D. When weights are initialized to zero

Q9. (Multiple Choice) [Hard]

In CNN architecture, what does a Pooling Layer do?

- A. Adds more features to the feature map
- B. Reduces the spatial dimensions of the feature map while retaining important features
- C. Applies activation functions to neurons
- D. Connects all neurons in one layer to all neurons in the next

Q10. (Multiple Choice) [Hard]

What is the difference between a shallow neural network and a deep neural network?

- A. Shallow has 1 hidden layer; Deep has 2 or more hidden layers
- B. Shallow uses ReLU; Deep uses Sigmoid
- C. Shallow is for regression; Deep is for classification
- D. Shallow trains faster always; Deep trains slower always

Q11. (True / False) [Easy]

A Neural Network with zero hidden layers is equivalent to Logistic Regression.

- A. True
- B. False

Q12. (True / False) [Medium]

Increasing the number of layers always improves a Neural Network's performance.

- A. True
- B. False

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Q13. (True / False) [Medium]

CNNs are the standard architecture for medical image classification tasks.

- A. True B. False

Q14. (True / False) [Hard]

Batch Normalization speeds up training by normalizing inputs to each layer.

- A. True B. False

Q15. (Short Answer) [Easy]

Explain how a Neural Network learns with a simple analogy.

Q16. (Short Answer) [Medium]

What are the main components of a CNN architecture? Explain each briefly.

Q17. (Short Answer) [Hard]

Explain Transfer Learning and why it is especially useful in biomedical AI.

Q18. (Multiple Choice) [Easy]

What activation function is used in the output layer for binary classification?

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- A. ReLU
- B. Sigmoid
- C. Softmax
- D. Tanh

Q19. (Multiple Choice) [Medium]

What is an Epoch in Neural Network training?

- A. One single weight update
- B. One complete pass through the entire training dataset
- C. One layer of the neural network
- D. One batch of training samples

Q20. (Short Answer) [Hard]

How does a CNN detect diseases in medical images? Explain with reference to chest X-ray analysis.
